ASSIGNMENT 1

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Question 1: Write Java Program to Check Leap Year

```java
import java.util.*;

public class leap{
    public static void main(String[] args){
        Scanner sc = new Scanner (System.in);
        System.out.println("Enter a Year:");
        int a=sc.nextInt();
        if((a%400==0)||(a%4==0&&a%100==0))
         System.out.print("Leap Year");
        else
         System.out.print("Not a leap year");
    }
}
```

```
Enter a Year:
2020
Not a leap year
=== Code Execution Successful ===
```

2. Write Java Program to Check Whether a Character is Alphabet or Not

```java
import java.util.*;

public class leap{
    public static void main(String[] args){
        Scanner sc = new Scanner (System.in);
        System.out.println("Enter a character:");
        char ch=sc.next().charAt(0);
          if ((ch >= 'A' && ch <= 'Z') || (ch >= 'a' &&
              ch <= 'z'))
           System.out.println("It is an alphabet");
        else
           System.out.println("It is not an alphabet");
    }
}
```

```
Enter a character:
F
It is an alphabet

=== Code Execution Successful ===
```

3.Write Java Program to Find Factorial of a Number

```java
import java.util.*;

public class factorial {
    public static void main(String[] args){
        Scanner sc = new Scanner (System.in);
        System.out.println("Enter a number:");
        int a= sc.nextInt();
        int fact=1;
        for(int i=1;i<=a;i++){
            fact=fact*i;

        }
        System.out.println("Factorial is:"+fact);

}
}
```

```
Enter a number:
5
Factorial is:120

=== Code Execution Successful ===
```

4. Write Java Program to Display Fibonacci Series

```java
import java.util.Scanner;

public class Fibonacci {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of terms: ");
        int n = sc.nextInt();

        int first = 0, second = 1;

        for (int i = 1; i <= n; i++) {
            System.out.print(first + " ");
            int next = first + second;
            first = second;
            second = next;
        }

        sc.close();
    }
}
```

```
Enter the number of terms: 6
0 1 1 2 3 5
=== Code Execution Successful ===
```

5. Write Java Program to Find GCD of two Numbers

```java
import java.util.Scanner;

public class GCD {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int a = sc.nextInt();

        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        int gcd = 1;

        for (int i = 1; i <= a && i <= b; i++) {
            if (a % i == 0 && b % i == 0) {
                gcd = i;
            }
        }

        System.out.println("GCD = " + gcd);

        sc.close();
    }
}
```

```
Enter first number: 16
Enter second number: 32
GCD = 16

=== Code Execution Successful ===
```

6. Write Java Program to Find LCM of two Numbers

```java
import java.util.Scanner;

public class LCM {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int a = sc.nextInt();

        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        int max = (a > b) ? a : b;

        while (true) {
            if (max % a == 0 && max % b == 0) {
                System.out.println("LCM = " + max);
                break;
            }
            max++;
        }

        sc.close();
    }
}
```

```
Enter first number: 12
Enter second number: 24
LCM = 24

=== Code Execution Successful ===
```

7. Write Java Program to Count Number of Digits in an Integer

```java
import java.util.Scanner;

public class number{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter  number: ");
        int a = sc.nextInt();
        int n=0;
        if (a==0){
            n=1;
        }
        else{
        while(a!=0){
            n++;
            a=a/10;
        }
        }
        System.out.print("no. of digits:"+n);

        sc.close();
    }
}
```

```
Enter  number: 678512
no. of digits:6
=== Code Execution Successful ===
```

8. Write Java Program to Reverse a Number

```java
import java.util.Scanner;

public class number{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter  number: ");
        int a = sc.nextInt();
        int n=0;

           do{
               int r= a%10;
               n=n*10+r;
               a=a/10;
           } while(a!=0);

        System.out.print(n);
    }
}
```

```
Enter  number: 2345
5432
=== Code Execution Successful ===
```

9. Write Java Program to Calculate the Power of a Number

```java
import java.util.Scanner;

public class number{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter  number: ");
        int a = sc.nextInt();
        System.out.print("Enter  power: ");
        int p= sc.nextInt();
        int n=1;
        for(int i =1; i<=p; i++){
            n=n*a;
        }
        System.out.print(n);
    }
}
```

```
Enter  number: 10
Enter  power: 3
1000
=== Code Execution Successful ===
```

10. Write Java Program to Check Palindrome

```java
import java.util.Scanner;

public class Palindrome {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        int original = num;
        int reverse = 0;

        while (num != 0) {
            int digit = num % 10;
            reverse = reverse * 10 + digit;
            num = num / 10;
        }

        if (original == reverse)
            System.out.println("Palindrome");
        else
            System.out.println("Not a Palindrome");

        sc.close();
    }
}
```

```
Enter a number: 1234
Not a Palindrome

=== Code Execution Successful ===
```